

Internship Project Report

House Price Prediction using Linear Regression

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Internship: **Artificial Intelligence & Machine Learning**

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Table of Contents

Table of Contents	2
1. Introduction	2
2. Dataset & Tools	2
3. Machine Learning Workflow	2
4. Exploratory Data Analysis (EDA)	3
5. Methodology & Model Training	3
6. Evaluation & Results	3
7. Conclusion & Future Scope	4
8. References	4

1. Introduction

This project aims to build a Multiple Linear Regression model to predict California house prices based on demographic and geographic features. The objective is to demonstrate an end-to-end Machine Learning pipeline, achieving reasonable accuracy in estimating median house values.

2. Dataset & Tools

Dataset: California Housing Dataset (20,640 samples, 8 features including Median Income, House Age, Rooms, Population, Coordinates). Target: Median House Value.

Tools & Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn.

3. Machine Learning Workflow

Data Collection → Data Preprocessing → Exploratory Data Analysis → Train-Test Split → Linear Regression Modeling → Prediction & Evaluation.

4. Exploratory Data Analysis (EDA)

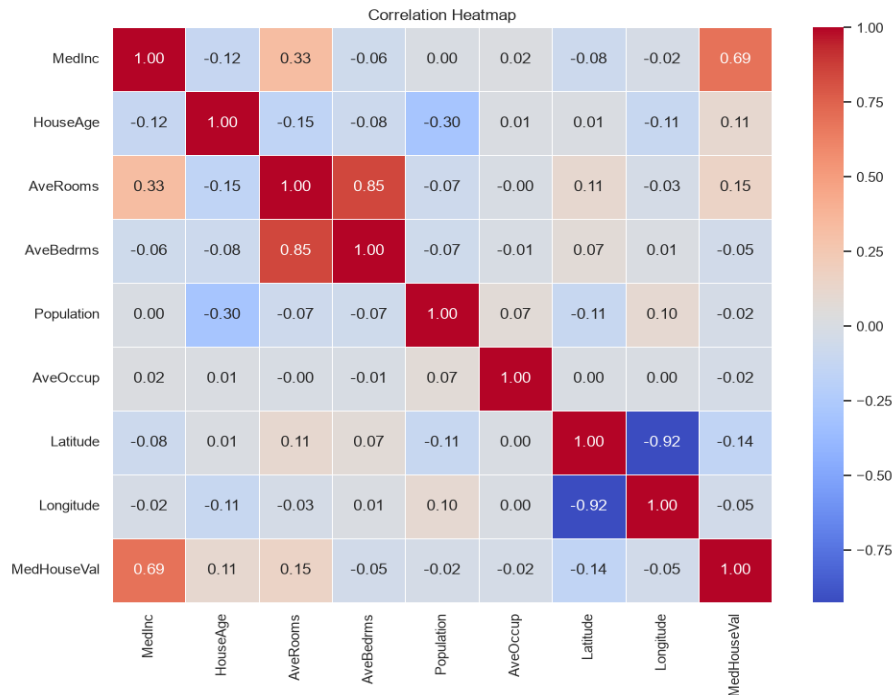


Figure 1: Correlation Heatmap. Displays feature relationships, highlighting Median Income's strong positive correlation with house values.

5. Methodology & Model Training

Data was split into training (80%) and testing (20%) sets. A Multiple Linear Regression model was implemented using Scikit-learn to identify linear relationships and predict target values.

6. Evaluation & Results

Metric	Value
Mean Absolute Error (MAE)	0.5332
Mean Squared Error (MSE)	0.5559
Root Mean Squared Error (RMSE)	0.7456
R-squared (R2) Score	0.5758

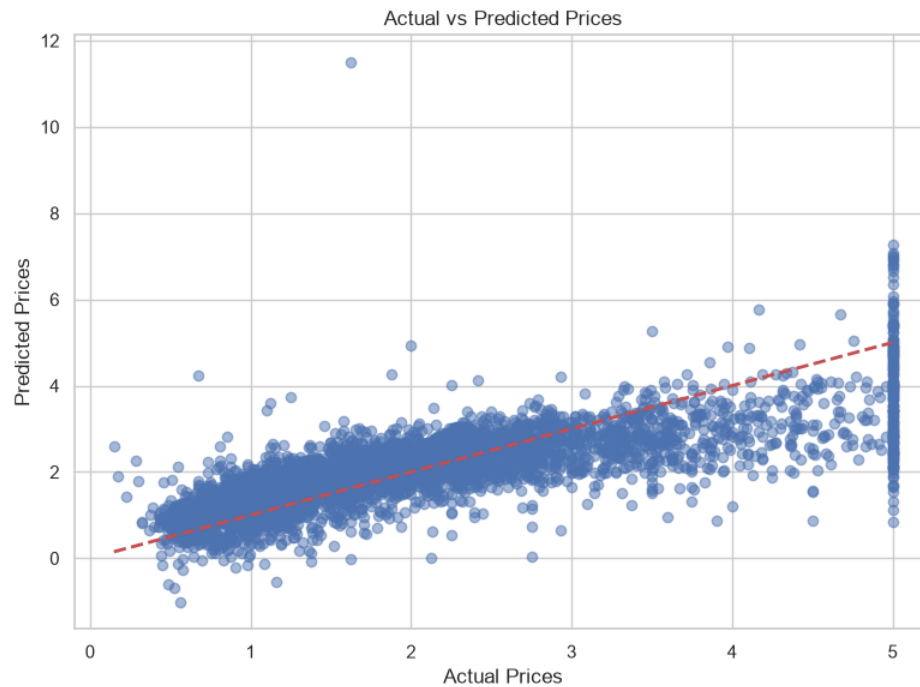


Figure 2: Actual vs Predicted Prices. Points closer to the diagonal line represent more accurate predictions by the model.

7. Conclusion & Future Scope

The Linear Regression model effectively captures general pricing trends. Future work could explore advanced ensemble methods like Random Forests and perform feature engineering to further improve predictive accuracy.

8. References

- Scikit-Learn Documentation: <https://scikit-learn.org/>
- California Housing Dataset documentation and resources.
- Python, Pandas, Matplotlib, and Seaborn official documentation.